ADMM-HFNet: A Matrix Decomposition-Based Deep Approach for Hyperspectral Image Fusion

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***Abstract*— Hyperspectral image (HSI) fusion refers to the reconstruction of a high-resolution HSI by fusing a low-resolution HSI (LR-HSI) and a high-resolution multispectral image (HR-MSI) over the same scene. Recently, researchers have proposed many approaches to handle this issue. However, most of them assume that both the spatial and spectral degradation functions are known, which are often limited or unavailable in reality. This article presents a novel model-driven deep network based on matrix decomposition, which considers spectral correlations and reasonably embeds the well-known observation models. Specifically, the proposed method decomposes the desired HSI into spectral basis and coefficients. The spectral basis can be estimated from the LR-HSI via singular value decomposition. To learn the coefficients, a learning model is constructed by merging the observation models, matrix decomposition, and sparsity into a concise single formulation. For solving the proposed model, a deep framework is built by unrolling the alternating direction method of multipliers (ADMM), dubbed as ADMM-HFNet, where the involved parameters can be learned adaptively. It is worth noting that the spectral basis cannot fully represent the desired HSI. Therefore, another model is constructed here to supplement the approximation error, which can also be embedded in the deep network. After checking on three datasets, it is found that the proposed method stands out from advanced competing techniques in both quality measures and visual effects.**

***Index Terms*— Alternating direction method of multipliers (ADMM), hyperspectral image (HSI) fusion, image reconstruction, matrix decomposition, model-driven deep learning.**

I. Introduction

HYPERSPECTRAL images (HSIs) are full of abundant spectral information, which can provide correct knowledge about ground substances and enhance the performance in different applications, including image classification [1]–[5], object detection [6]–[8], tracking [9]–[11], and segmentation [12]–[15]. However, the current imaging equipment restricts the coexistence of high spatial resolution and high spectral resolution. As a result, HSIs usually contain more spectral information and lesser spatial information than multispectral images (MSIs). The limited spatial resolution is not conducive to the application and development of HSIs. Therefore, it is necessary to improve the spatial resolution in HSI imaging. For this, one way is to improve the physical resolution of the hardware. It may increase the volume and weight of the device, which is not beneficial for satellite remote sensing imaging and may lead to higher costs. The other way for improving the spatial resolution in HSI imaging is image fusion, which generates the desired HSI by fusing a high-resolution MSI (HR-MSI) and a low-resolution HSI (LR-HSI) through a set of algorithms. The image fusion is a practical technique as the LR-HSI and HR-MSI possess sufficient spectral and spatial information over the same scene, respectively. Although the image fusion is an ill-posed inverse problem where the desired image cannot be reconstructed completely, we can approximate it by using the available information. A lot of advanced methods have been put forward for the fusion task.

Traditional methods usually encode the prior structure underlying the desired HSI into a regularizer for solving the fusion problem. Various prior structures are considered, such as sparse representation [16], [17], spatial smoothness [18], [19], and spectral correlation [20]–[23]. Usually, these methods are algorithmically interpretable and can make full use of the well-known observation models. However, the inherent complex spatial and spectral structures of the HSIs cannot be reflected fully and adaptively. Unlike traditional methods, deep learning-based (DL-B) methods require less prior knowledge and can achieve better performance due to their strong learning capacity and fast parallel computing. However, the adopted network architectures are usually migrated from other fields, such as classification and recognition, which are not tailor-made for image fusion. Besides, the observation models and some useful prior knowledge are not considered in current DL-B methods. Consequently, the dependence between the images and latent prior structure cannot be well reflected in the network, and its fusion accuracy is also hindered.

In order to deal with the aforementioned issues, we put forward a promising deep unfolding network for bridging the gap between traditional and DL-B methods with good ability for interpretation and generalization. On the one hand,

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similar to traditional methods, the proposed network utilizes the observation models and the correlations between spectral bands. On the other hand, similar to DL-B methods, it can be trained to learn the involved parameters adaptively for improving effectiveness and efficiency. To achieve these, the desired HSI is decomposed into the spectral basis and the coefficients following the hyperspectral unmixing model. Considering that the LR-HSI shares almost the same spectral information as the desired HSI, the spectral basis is learned via the singular value decomposition (SVD) algorithm. To learn the coefficients, we construct a learning model by merging the observation models, matrix decomposition, and sparsity into a single concise formulation. The alternating direction method of multipliers (ADMM) can be utilized to address this learning model. Since it is difficult to tune the regularization parameters and the involved spectral response function (SRF) and point spread function (PSF) are also usually unknown in practice, we consider the use of the deep CNN to approximate the model so that the regularization parameters as well as the SRF- and PSF-related parameters can be learned adaptively. Therefore, a deep network is built by unfolding the ADMM algorithm for solving the proposed model, named ADMM-HFNet. However, the desired HSI cannot be fully represented by such spectral basis and coefficients. Hence, we construct an information supplement model to learn the approximate error, which utilizes the observation models and the super-resolution image synthesized by the spectral basis and coefficients. This model can also be embedded in the deep unfolding network. For further improvement of the performance, the fusion results are obtained by concatenating the output of each stage, so as to employ multilevel information. In comparison to the recent advanced fusion methods, the proposed method shows superior performance.

The main contributions of the present research are summarized as follows.

1) A matrix decomposition-based model-driven deep learning approach is proposed for the first time for performing fusion. This method considers the correlations between spectral bands and the powerful learning capability of deep CNNs.
2) We transform the ADMM iterations into a deep network so that the regularization parameters and the SRF- and PSF-related parameters can be learned adaptively through end-to-end training. This is very beneficial for the fusion of hyperspectral real data.
3) An information supplement model is constructed to learn such information, which cannot be learned from the dynamic matrix decomposition. It can also be unrolled and embedded in a deep network.
4) The hierarchical sampling structure designed in the proposed ADMM-HFNet can handle HSI fusion of arbitrary resolution ratios including noninteger ones, which facilitates the application of real fusion.

The rest of this article is organized as follows. Section II surveys the related works about the fusion task. In Section III, the proposed ADMM-HFNet approach is presented in detail. In Section IV, the experimental results and corresponding discussion are provided. Conclusions and future work are drawn in Section V. To facilitate the description of this article, the significant notations appeared in the formulas are shown in Table I.

TABLE I
DESCRIPTION OF THE SIGNIFICANT NOTATIONS IN THE FORMULAS

Notations	Description
$\|\cdot\|_F$	Frobenius norm
$\|\cdot\|_1$	L1-norm
$\Delta(\cdot)$	Derivative
$\lfloor\cdot\rfloor$	Round-down operation
$\mathbf{X}$	Matrix form of variable
$\mathcal{X}$	Tensor form of variable
a_0	Regularization parameter
$\mathcal{D}_0$	Intermediate variable
$\mathcal{X} \times_3 \mathbf{R}$	The third-mode multiplication of tensor and matrix
$\mathbf{X}^{(k)}$	The k-th iteration of $\mathbf{X}$
$\mathbf{X}^{(k,i)}$	The i-th inner iteration in the k-th iteration of $\mathbf{X}$
svds	SVD function
soft	Soft thresholding function
sign	Signum function
$\downarrow^r_{\theta(k)}(\cdot)$	Downsample operation with ratio r and parameters θ in k-th stage
$\uparrow^r_{\theta(k)}(\cdot)$	Upsample operation with ratio r and parameters θ in k-th stage

II. RELATED WORK

In this section, we divide the current HSI fusion methods roughly into three categories: traditional methods [24]–[35], DL-B methods [36]–[45], and model-driven DL-B methods [46], [47].

A. Traditional Methods

Traditional methods usually construct an optimization model for solving the fusion problem by combining the observation models and some manual priors. For example, Palsson *et al.* [24] adopted the total variation (TV) regularization approach to remove noise and preserved edges. Taking spectral correlations into account, matrix decomposition-based methods were proposed to improve the performance by reducing spectral distortions. Yokoya *et al.* [25] represented HSI with endmembers and abundances and learned them from LR-HSI and HR-MSI, respectively, based on the nonnegative spectral unmixing. Taking the sparsity prior into account, Dong *et al.* put forward a nonnegative structured sparse coding method to employ both the spectral and spatial correlations based on the clustering algorithm. As tensor decomposition can better protect the image structure and further exploit the spectral–spatial correlations, Li *et al.* [31] used a sparse core tensor and the three-mode dictionaries to represent an HSI cube according to the Tucker decomposition. Xu *et al.* [34] put forward a nonlocal decomposition model through the coupled tensor canonical polyadic decomposition. The factor matrices can be shared between HR-MSI and the desired HSI.

B. DL-B Methods

Due to the strong learning capacity of deep CNN, DL-B methods could be applied successfully to many

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visual processing fields, including the HSI fusion. Various promising network architectures have also been proposed. Masi *et al.* [36] first stacked the interpolated LR-HSI and the panchromatic image in the spectral dimension and then sent them to a three-layer CNN for learning a map. Considering that the spatial–spectral information is difficult to extract, Palsson *et al.* [37] proposed a full 3-D CNN to address the fusion task. Assuming that the spatial and spectral information can be separated, Yang *et al.* [38] proposed a two-branch network to extract and fuse spatial and spectral features. Zhou *et al.* [39] proposed a pyramid network with two subnetworks to make the best of the multiscale spatial information. To focus on important information by neglecting useless information, Zheng *et al.* [41] put forward a dual-attention network, which is made up of channel attention and spatial attention.

C. Model-Driven DL-B Methods

Model-driven DL-B methods [48]–[52] usually transform the iterative solver of an optimization model into a deep network, which takes the advantages of both traditional and DL-B methods. A deep network is not a black box. It becomes task-oriented where the modules have some physical meanings. Optimization models can be solved efficiently in the deep network. As a result, embedded models guide the network to the optimal solution and achieve higher accuracy and faster convergence speed. Recently, these types of methods have been applied to the HSI fusion task and achieved the outstanding performance. For example, Xie *et al.* [46] combined an observation model related to LR-HSI and low-rankness prior knowledge to construct an optimization model. They unfolded the proximal gradient algorithm in a deep network to solve the model. Wang *et al.* [47] developed a model to update the desired HSI by fusing the backprojection errors. The iterative procedure is unfolded into a network, where the observation models can be estimated, and the fusion model can be learned iteratively and adaptively. Our method belongs to this research line, which unfolds the solvers of fusion models into a deep network. The network becomes more interpretable, and the models can be better solved under the powerful learning circumstance.

III. Proposed Method

In this section, we first present the models and solvers of the proposed ADMM-HFNet and then unroll the solvers into a deep architecture.

A. Problem Formulation

The desired HR-HSI $\mathbf{X} \in \mathbb{R}^{S\times HW}$ can be estimated by fusing the LR-HSI $\mathbf{Y} \in \mathbb{R}^{S\times hw}$ and HR-MSI $\mathbf{Z} \in \mathbb{R}^{s\times HW}$ over the same scene, where S and s are the numbers of bands ($s \ll S$), H and h are the heights ($h \ll H$) of the images, and W and w are the widths ($w \ll W$) of the images.

$\mathbf{Y}$ and $\mathbf{Z}$ are, respectively, the spatially and spectrally downsampled products of $\mathbf{X}$. According to the generation mechanism, the observation models can be described as

$$\mathbf{Y} = \mathbf{XC} + \mathbf{N}_h \tag{1}$$

$$\mathbf{Z} = \mathbf{RX} + \mathbf{N}_m \tag{2}$$

where $\mathbf{C} \in \mathbb{R}^{HW\times hw}$ denotes the spatial degradation operation, which is often made up of a blurring matrix and a downsampling matrix. $\mathbf{R} \in \mathbb{R}^{s\times S}$ denotes the spectral downsampling matrix, and $\mathbf{N}_h$ and $\mathbf{N}_m$ denote the noise in the observation models.

B. Estimation of Spectral Basis

Hyperspectral data usually have spectral correlations, and spectral signatures usually live in a relatively lower dimensional space. We can decompose HR-HSI as

$$\mathbf{X} = \mathbf{PA} + \mathbf{E} \tag{3}$$

where $\mathbf{P} \in \mathbb{R}^{S\times c}$ and $\mathbf{A} \in \mathbb{R}^{c\times HW}$ denote the spectral basis and coefficients, respectively. Small values of c ($c < S$) represent a low rank, and $\mathbf{E}$ is the approximation error.

This matrix decomposition has three major advantages. First, the high correlations among spectral bands are fully exploited, which can help to improve the fusion performance. Second, working in a lower dimensional space can reduce the computational complexity. Third, the performance can be improved by removing noise via dimensionality reduction.

Based on the matrix decomposition, the fusion goal becomes the learning of the spectral basis $\mathbf{P}$ and coefficients $\mathbf{A}$. Consider that $\mathbf{X}$ and $\mathbf{Y}$ are spectrally correlated, and they share almost the same spectral information. Consequently, we can learn $\mathbf{P}$ from $\mathbf{Y}$ via SVD as follows:

$$\mathbf{U}, \Sigma, \mathbf{P}^T = \text{svds}(\mathbf{Y}^T, c) \tag{4}$$

where $\mathbf{U} \in \mathbb{R}^{HW\times c}$ denotes a semiunitary matrix, $\Sigma \in \mathbb{R}^{c\times c}$ denotes a diagonal matrix containing the singular values in nonincreasing order, and svds is the SVD function reserving the top c largest singular values for dimensionality reduction.

C. Estimation of Coefficients

Keeping $\mathbf{P}$ fixed, coefficients $\mathbf{A}$ can be optimized following (1)–(3). Sparsity prior has shown promising performance by solving such ill-posed inverse problems under HSI reconstruction circumstances. Here, we assume that only a few distinct materials can be presented at each pixel location x_i of the desired HSI $\mathbf{X}$. Therefore, the coefficient matrix $\mathbf{A}$ is sparse. Taking the sparsity of coefficients into consideration and replacing the L0-norm with L1-norm under approximate equivalent conditions, the estimation model can be formulated as follows:

$$\min_{\mathbf{A}} \|\mathbf{RPA} - \mathbf{Z}\|_F^2 + \lambda\|\mathbf{PAC} - \mathbf{Y}\|_F^2 + \mu\|\mathbf{A}\|_1 \tag{5}$$

where $\|\cdot\|_F$ and $\|\cdot\|_1$ denote the Frobenius norm and L1-norm, respectively, λ is a parameter used to balance the weight occupied by the spatial and spectral errors, and μ is the penalty parameter imposed on the solution sparsity.

The ADMM technique can be used to solve (5). By bringing splitting variables $\mathbf{V} = \mathbf{A}$ and $\mathbf{T} = \mathbf{PV}$, we can get the

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augmented Lagrangian function as

$$\begin{aligned}&\mathcal{L}(\mathbf{A},\mathbf{V},\mathbf{T},\mathbf{G},\mathbf{H})\\&=\|\mathbf{RPV}-\mathbf{Z}\|_F^2+\lambda\|\mathbf{TC}-\mathbf{Y}\|_F^2\\&\quad+\mu\|\mathbf{A}\|_1+\eta_1\|\mathbf{A}-\mathbf{V}+\frac{\mathbf{G}}{2\eta_1}\Big\|_F^2+\eta_2\|\mathbf{PV}-\mathbf{T}+\frac{\mathbf{H}}{2\eta_2}\Big\|_F^2\end{aligned}\tag{6}$$

where η_1 and η_2 are two penalty parameters and $\mathbf{G}$ and $\mathbf{H}$ are the corresponding Lagrangian multipliers. The above function can be transformed into the optimization of the following three subproblems:

$$\begin{aligned}\min_{\mathbf{V}}\ &\|\mathbf{RPV}-\mathbf{Z}\|_F^2+\eta_1\left\|\mathbf{A}^{(k)}-\mathbf{V}+\frac{\mathbf{G}^{(k)}}{2\eta_1}\right\|_F^2\\&+\eta_2\left\|\mathbf{PV}-\mathbf{T}^{(k)}+\frac{\mathbf{H}^{(k)}}{2\eta_2}\right\|_F^2\end{aligned}\tag{7}$$

$$\min_{\mathbf{T}}\ \lambda\|\mathbf{TC}-\mathbf{Y}\|_F^2+\eta_2\left\|\mathbf{PV}^{(k+1)}-\mathbf{T}+\frac{\mathbf{H}^{(k)}}{2\eta_2}\right\|_F^2\tag{8}$$

$$\min_{\mathbf{A}}\ \mu\|\mathbf{A}\|_1+\eta_1\left\|\mathbf{A}-\mathbf{V}^{(k+1)}+\frac{\mathbf{G}^{(k)}}{2\eta_1}\right\|_F^2\tag{9}$$

where k is the iteration counter.

The Lagrangian multipliers are updated as follows:

$$\mathbf{G}^{(k+1)}=\mathbf{G}^k+d_0(\mathbf{A}^{(k+1)}-\mathbf{V}^{(k+1)})\tag{10}$$

$$\mathbf{H}^{(k+1)}=\mathbf{H}^k+d_1(\mathbf{PV}^{(k+1)}-\mathbf{T}^{(k+1)}).\tag{11}$$

where $d_0=2\eta_1$ and $d_1=2\eta_2$.

It is difficult to directly calculate the inverse of matrices in closed-form solutions of problems (7) and (8). Therefore, we employ the gradient descent algorithm (GDA) to approximate the solutions. The derivations are

$$\begin{aligned}\Delta(\mathbf{V})=&\,2(\mathbf{RP})^T(\mathbf{RPV}-\mathbf{Z})-2\eta_1\left(\mathbf{A}-\mathbf{V}+\frac{\mathbf{G}^{(k)}}{2\eta_1}\right)\\&+2\eta_2\mathbf{P}^T\left(\mathbf{PV}-\mathbf{T}^{(k)}+\frac{\mathbf{H}^{(k)}}{2\eta_2}\right)\end{aligned}\tag{12}$$

$$\Delta(\mathbf{T})=2\lambda(\mathbf{TC}-\mathbf{Y})\mathbf{C}^T-2\eta_2\left(\mathbf{PV}^{(k+1)}-\mathbf{T}+\frac{\mathbf{H}^{(k)}}{2\eta_2}\right).\tag{13}$$

The solutions can be approximated as

$$\mathbf{V}^{(k,i)}=\mathbf{V}^{(k,i-1)}-l_1\Delta(\mathbf{V}^{(k,i-1)})\tag{14}$$

$$\mathbf{T}^{(k,j)}=\mathbf{T}^{(k,j-1)}-l_2\Delta(\mathbf{T}^{(k,j-1)})\tag{15}$$

where $i\in\{1,\ldots,K_1\}$ and $j\in\{1,\ldots,K_2\}$ indicate, respectively, the ith and jth inner iterations and l_1 and l_2 are the step sizes in the gradient descent method. Therefore, the solutions are formulated as follows:

$$\begin{aligned}\mathbf{V}^{(k,i)}=&\,a_0\mathbf{V}^{(k,i-1)}-a_1\mathbf{P}^T\mathbf{R}^T(\mathbf{RPV}^{(k,i-1)}-\mathbf{Z})\\&+a_2\mathbf{A}^{(k)}+a_3\mathbf{G}^{(k)}+a_4\mathbf{P}^T(\mathbf{T}^{(k)}-a_5\mathbf{H}^{(k)})\end{aligned}\tag{16}$$

$$\begin{aligned}\mathbf{T}^{(k,j)}=&\,b_0\mathbf{T}^{(k,j-1)}-b_1(\mathbf{T}^{(k,j-1)}\mathbf{C}-\mathbf{Y})\mathbf{C}^T\\&+b_2\mathbf{PV}^{(k+1)}+b_3\mathbf{H}^{(k)}\end{aligned}\tag{17}$$

where $\mathbf{P}$ is semiunitary ($\mathbf{P^TP}=\mathbf{I}$), $a_0=1-2l_1\eta_1-2l_1\eta_2$, $a_1=2l_1$, $a_2=2l_1\eta_1$, $a_3=l_1$, $a_4=2l_1\eta_2$, $a_5=1/2\eta_2$, $b_0=1-2l_2\eta_2$, $b_1=2l_2\lambda$, $b_2=2l_2\eta_2$, and $b_3=l_2$. In addition to those, $\mathbf{V}^{(k,0)}$ is initialized as $\mathbf{V}^{(k)}$ whose updated result is $\mathbf{V}^{(k+1)}=\mathbf{V}^{(k,K_1)}$ and $\mathbf{T}^{(k,0)}$ is initialized as $\mathbf{T}^{(k)}$ whose updated result is $\mathbf{T}^{(k+1)}=\mathbf{T}^{(k,K_2)}$.

The solution of (9) is

$$\mathbf{A}^{(k+1)}=\text{soft}(\mathbf{V}^{(k+1)}-c_0\mathbf{G}^{(k)},c_1)\tag{18}$$

where $c_0=1/2\eta_1$, $c_1=\mu/2\eta_1$, and soft denotes the soft thresholding function that is calculated as follows:

$$\text{soft}(\mathbf{x},\delta)=\text{sign}(\mathbf{x})\times\max(|\mathbf{x}|-\delta,0).\tag{19}$$

D. Information Supplement Model

According to (3), although we get the estimated $\mathbf{P}$ and $\mathbf{A}$, the desired HSI $\mathbf{X}$ cannot be fully represented as the approximation error still exists there. Therefore, provision should be there to calculate such approximation error, which cannot be included in the linear unmixing model. To achieve this, Han *et al.* [53] represented the main information and residual information separately by using a spectral dictionary and a spatial dictionary, respectively. Instead of calculating the approximation error directly, we obtain the desired HSI by addressing the following optimization problem:

$$\begin{aligned}\min_{\mathbf{X}}\ &\|\mathbf{X}^{(k)}-\mathbf{PA}^{(k)}\|_F^2+\alpha\|\mathbf{X}^{(k)}-\mathbf{PV}^{(k)}\|_F^2\\&+\beta\|\mathbf{X}^{(k)}\mathbf{C}-\mathbf{Y}\|_F^2+\gamma\|\mathbf{RX}^{(k)}-\mathbf{Z}\|_F^2\end{aligned}\tag{20}$$

where α, β, and γ are three tradeoff parameters. The above problem could be solved as a Sylvester equation, but it does not work well in our latter unfolding network. Therefore, we still employ GDA to approximate the solution. The derivation of (20) with respect to $\mathbf{X}$ is

$$\begin{aligned}\Delta(\mathbf{X})=&\,2(\mathbf{X}-\mathbf{PA}^{(k)})+2\alpha(\mathbf{X}-\mathbf{PV}^{(k)})\\&+2\beta(\mathbf{XC}-\mathbf{Y})\mathbf{C}^T+2\gamma\,\mathbf{R}^T(\mathbf{RX}-\mathbf{Z}).\end{aligned}\tag{21}$$

The solution can be approximated as follows:

$$\mathbf{X}^{(k,i)}=\mathbf{X}^{(k,i-1)}-l_3\Delta(\mathbf{X}^{(k,i-1)})\tag{22}$$

where $i\in\{1,\ldots,K_3\}$ and l_3 is the step size in the gradient descent method. Therefore, the module is formulated as

$$\begin{aligned}\mathbf{X}^{(k,i)}=&\,e_0\mathbf{X}^{(k,i-1)}+e_1\mathbf{PA}^{(k)}+e_2\mathbf{PV}^{(k)}\\&-e_3(\mathbf{B}_1-\mathbf{Y})\mathbf{C}^T-e_4\mathbf{R}^T(\mathbf{B}_2-\mathbf{Z})\end{aligned}\tag{23}$$

where $\mathbf{B}_1=\mathbf{X}^{(k,i-1)}\mathbf{C}$, $\mathbf{B}_2=\mathbf{RX}^{(k,i-1)}$, $e_0=1-2l_3-2l_3\alpha$, $e_1=2l_3$, $e_2=2l_3\alpha$, $e_3=2l_3\beta$, and $e_4=2l_3\gamma$. The initial setting and the updated result are set as $\mathbf{X}^{(k,0)}=\mathbf{T}^{(k)}$ and $\mathbf{X}^{(k)}=\mathbf{X}^{(k,K_3)}$, respectively.

The desired HSI could be reconstructed analytically by executing Algorithm 1 for K number of iterations. However, two problems still exist there. The first problem is that the SRF and PSF must be known, which is limited or unavailable in practice. The other problem is the difficulty in tuning the involved parameters. To overcome these difficulties, we unfold the solvers and build a model-driven deep framework to complete the fusion task.

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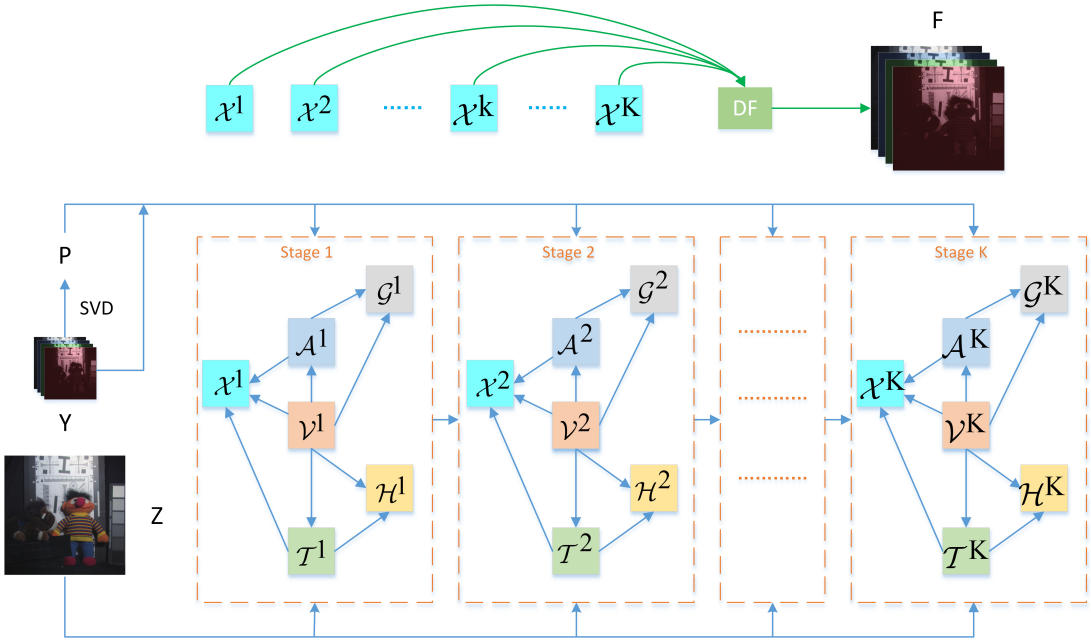

Fig. 1. Network structure of the proposed ADMM-HFNet. Top: reconstruction network using DF. Bottom: data flow graph with K stages.

Algorithm 1 ADMM-Based LR-HSI and HR-MSI Fusion

Input: $\mathbf{Y}$, $\mathbf{Z}$
Estimate the spectral basis $\mathbf{P}$ via (4);
for $k = 1 : K$ **do**
  Update $\mathbf{V}$ via (16);
  Update $\mathbf{T}$ via (17);
  Update $\mathbf{A}$ via (18);
  Update $\mathbf{G}$ via (10);
  Update $\mathbf{H}$ via (11);
  Update $\mathbf{X}$ via (23);
end for
Output: $\mathbf{X}$.

E. Deep Unfolding Network

Since the iterative solvers (16), (17), and (23) contain both inner and outer iterations, we simplify them before designing the unfolded network. Yang *et al.* [51] illustrated that the inclusion of more outer iterations in the updating block of auxiliary variables is more effective than adding more inner iterations. Hence, for simplicity, we empirically set the numbers of inner iterations K_1, K_2, and K_3 to 1. As a result, these three equations can be rewritten as follows:

$$\begin{aligned}\mathbf{V}^{(k+1)} &= a_0\mathbf{V}^{(k)} - a_1\mathbf{P}^T\mathbf{R}^T(\mathbf{RPV}^{(k)} - \mathbf{Z}) + a_2\mathbf{A}^{(k)} \\ &\quad + a_3\mathbf{G}^{(k)} + a_4\mathbf{P}^T(\mathbf{T}^{(k)} - a_5\mathbf{H}^{(k)}) \qquad (24)\\ \mathbf{T}^{(k+1)} &= b_0\mathbf{T}^{(k)} - b_1(\mathbf{T}^{(k)}\mathbf{C} - \mathbf{Y})\mathbf{C}^T \\ &\quad + b_2\mathbf{PV}^{(k+1)} + b_3\mathbf{H}^{(k)} \qquad (25)\\ \mathbf{X}^{(k+1)} &= e_0\mathbf{T}^{(k+1)} + e_1\mathbf{PA}^{(k+1)} + e_2\mathbf{PV}^{(k+1)} \\ &\quad - e_3(\mathbf{D}_1 - \mathbf{Y})\mathbf{C}^T - e_4\mathbf{R}^T(\mathbf{D}_2 - \mathbf{Z}) \qquad (26)\end{aligned}$$

where $\mathbf{D}_1 = \mathbf{T}^{(k+1)}\mathbf{C}$ and $\mathbf{D}_2 = \mathbf{RT}^{(k+1)}$.

By unfolding the aforementioned iterative solvers of models (5) and (20), we can develop a model-driven deep network. Such an unfolding technique is widely used in various computer vision tasks, whose effectiveness was substantiated in [46], [51], and [52]. The network structure of the proposed ADMM-HFNet is shown in Fig. 1, where the kth stage of the network corresponds to the kth iteration of the solvers. The subfigure at the bottom represents the data flow graph with K stages. Each of those stages consists of three data modules ($\mathcal{V}$, $\mathcal{T}$, and $\mathcal{A}$), two multiplier modules ($\mathcal{G}$ and $\mathcal{H}$), and an information supplement module ($\mathcal{X}$), which are expressed by (10), (11), (18), and (24)–(26). Each stage takes the LR-HSI $\mathbf{Y}$, HR-MSI $\mathbf{Z}$, spectral basis $\mathbf{P}$, and outputs of modules ($\mathcal{V}$, $\mathcal{T}$, $\mathcal{A}$, $\mathcal{G}$, and $\mathcal{H}$) of the previous stage as inputs and provides the updated results of the six modules as the outputs. The subfigure at the top represents the final reconstruction architecture with dense fusion (DF) [47], where the outputs of module $\mathcal{X}$ in all the stages are combined for reconstructing the final fusion image $\mathbf{F}$.

In order to protect the spatial structures of the images and also to facilitate the design of the network architectures, we use the variables with their tensor formats ($\mathcal{X} \in \mathbb{R}^{H\times W\times S}$, $\mathcal{Y} \in \mathbb{R}^{h\times w\times S}$, $\mathcal{Z} \in \mathbb{R}^{H\times W\times s}$, $\mathcal{V} \in \mathbb{R}^{H\times W\times c}$, $\mathcal{T} \in \mathbb{R}^{H\times W\times S}$, $\mathcal{A} \in \mathbb{R}^{H\times W\times c}$, $\mathcal{G} \in \mathbb{R}^{H\times W\times c}$, and $\mathcal{H} \in \mathbb{R}^{H\times W\times S}$). For convenience, a tensor is denoted by the same notation as the network module that updates the tensor. The design and detailed description of these modules are as follows.

1) Data Module $\mathcal{V}$: In the tensor version, (24) can be divided into the following fundamental operations:

$$\mathcal{D}_1{}^{(k)} = \mathcal{V}^{(k)} \times_3 \mathbf{P} \qquad (27)$$
$$\mathcal{D}_2{}^{(k)} = \mathcal{D}_1{}^{(k)} \times_3 \mathbf{R} - \mathcal{Z} \qquad (28)$$
$$\mathcal{D}_3{}^{(k)} = \mathcal{D}_2{}^{(k)} \times_3 \mathbf{R}^T \qquad (29)$$
$$\mathcal{D}_4{}^{(k)} = \mathcal{D}_3{}^{(k)} \times_3 \mathbf{P}^T \qquad (30)$$

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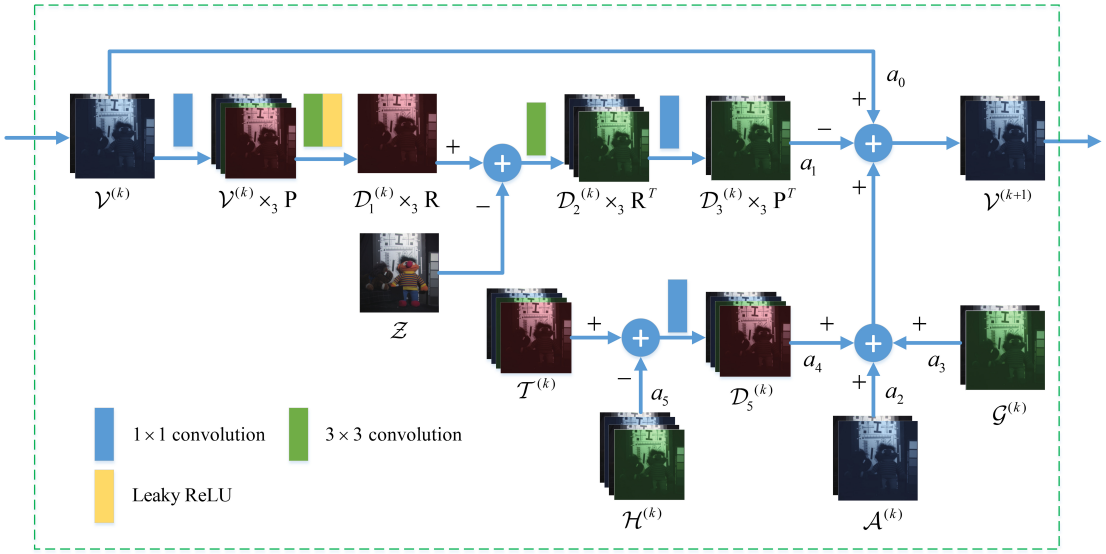

Fig. 2. Architecture of the module $\mathcal{V}$ at the kth stage.

$$\mathcal{D}_5{}^{(k)} = (\mathcal{T}^{(k)} - a_5\mathcal{H}^{(k)}) \times_3 \mathbf{P}^T \tag{31}$$

$$\mathcal{V}^{(k+1)} = a_0\mathcal{V}^{(k)} - a_1\mathcal{D}_4{}^{(k)} + a_2\mathcal{A}^{(k)} + a_3\mathcal{G}^{(k)} + a_4\mathcal{D}_5{}^{(k)} \tag{32}$$

where $\times_3$ denotes the third-mode multiplication for a tensor.

Accordingly, we design the module by performing the above operations. Equations (27)–(31) can be solved easily by using a 1×1 convolution as all of them contain a multiplication between a tensor and a matrix along the channel mode. Specifically, considering that the spectral basis $\mathbf{P}$ is known, we directly reshape it and its transpose into convolutional kernels of size $1 \times 1 \times c \times S$ and $1 \times 1 \times S \times c$, respectively. The spectral downsampling matrix $\mathbf{R}$ depicts the linear relationship between each band of HR-MSI and the relevant bands of HR-HSI. It usually needs to satisfy nonnegativity. Also, the sum of each row should be 1. Actually, these constraints are unfavorable to some extent for expressing the relationship between images as the observation models are not simply linear. Therefore, we drop these constraints from this unfolded network. For simulating $\mathbf{R}$ and its transpose, we define two learnable convolutional kernels of size $3 \times 3 \times S \times s$ and $3 \times 3 \times s \times S$, from which a better relationship expression can be obtained by using the local spatial information. Besides, we apply the leaky rectified linear unit (ReLU) as the activation function for increasing the nonlinearity of $\mathbf{R}$ after its implementation. Equation (32) gives a linear fusion structure for updating the temporary coefficients $\mathcal{V}^{(k)}$ by the linear representation of the coefficients $\mathcal{A}^{(k)}$, multiplier $\mathcal{G}^{(k)}$, ameliorative gradients $\mathcal{D}_4{}^{(k)}$ and $\mathcal{D}_5{}^{(k)}$, and $\mathcal{V}^{(k)}$ itself. The constant parameters a_0–a_5 can be learned adaptively in the network. The detailed architecture of the module $\mathcal{V}$ is shown in Fig. 2.

2) Data Module $\mathcal{T}$: Equation (25) can be divided into the following fundamental operations:

$$\mathcal{E}_1{}^{(k)} = \downarrow_{\theta^{(k)}}^{r}(\mathcal{T}^{(k)}) - \mathcal{Y} \tag{33}$$

$$\mathcal{E}_2{}^{(k)} = \uparrow_{\theta^{(k)}}^{r}(\mathcal{E}_1{}^{(k)}) \tag{34}$$

$$\mathcal{E}_3{}^{(k)} = \mathcal{V}^{(k+1)} \times_3 \mathbf{P} \tag{35}$$

$$\mathcal{T}^{(k+1)} = b_0\mathcal{T}^{(k)} - b_1\mathcal{E}_2{}^{(k)} + b_2\mathcal{E}_3{}^{(k)} + b_3\mathcal{H}^{(k)} \tag{36}$$

where $\downarrow_{\theta^{(k)}}^{r}(\cdot)$ and $\uparrow_{\theta^{(k)}}^{r}(\cdot)$ play the role of the downsampling block $\mathbf{C}$ and upsampling block $\mathbf{C}^T$, respectively, r denotes the resolution ratio, and $\theta^{(k)}$ denotes the parameters involved in the blocks at the kth stage of the network.

Equations (33) and (34) can be solved by a convolution and a deconvolution with the corresponding step size, respectively. For example, Wang *et al.* [47] modeled $\downarrow_{\theta^{(k)}}^{r}(\cdot)$ and $\uparrow_{\theta^{(k)}}^{r}(\cdot)$ by using, respectively, a 12×12 convolution and a deconvolution with a stride of r. Different from them, we decompose the sampling operations into multilayer structures for capturing richer features and depicting the spatial observation model in a better way. Besides, we consider the cases of noninteger resolution ratios so that the proposed method can handle the fusion of arbitrary ratios.

We divide the resolution ratio r ($r = H/h = W/w$) into factors r_1 and r_2, which are calculated as follows:

$$r_1 = \lfloor \sqrt{r} \rfloor \tag{37}$$

$$r_2 = r/r_1 \tag{38}$$

where $\lfloor \cdot \rfloor$ denotes the round-down operator. Obviously, r_1 is an integer, and r_2 may be a decimal. When r_2 is an integer, we perform $\downarrow_{\theta^{(k)}}^{r}(\cdot)$ by using two convolutions with kernel sizes of $(r_2 + 4) \times (r_2 + 4)$ and $(r_1 + 4) \times (r_1 + 4)$ and strides of r_2 and r_1, respectively. Similarly, $\uparrow_{\theta^{(k)}}^{r}(\cdot)$ is performed by using two deconvolutions with kernel sizes of $(r_1 + 4) \times (r_1 + 4)$ and $(r_2 + 4) \times (r_2 + 4)$, and strides of r_1 and r_2, respectively. The spatial sizes of the kernels are appropriately extended by 4 pixels for widening the receptive fields. When r_2 is a decimal, an interpolation algorithm needs to be adopted. Specifically, to perform $\downarrow_{\theta^{(k)}}^{r}(\cdot)$, we first scale the input of the block to an image with a spatial resolution of $(h \times r_1) \times (w \times r_1)$ by using a bilinear algorithm. Considering that the interpolation algorithm is static, we add a 5×5 convolution with a stride of 1 to enhance its ability of dynamic

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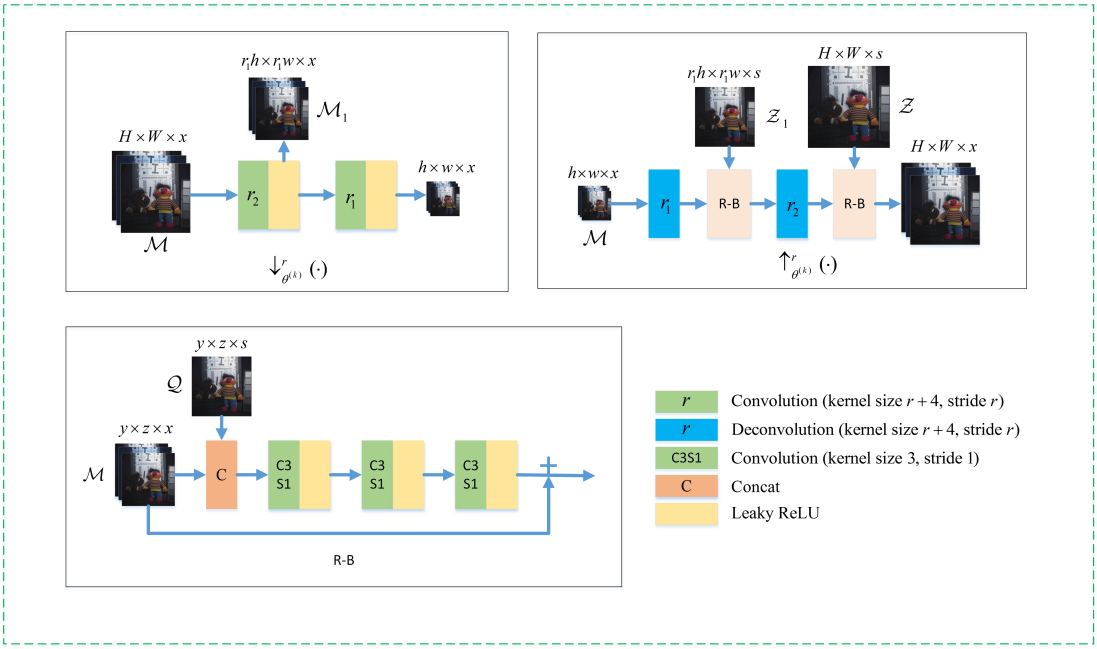

Fig. 3. Structures of the $\downarrow^r_{\theta^{(k)}}(\cdot)$ and $\uparrow^r_{\theta^{(k)}}(\cdot)$ over an integer factor r_2. $\mathcal{M}$ denotes the current input of each block and $\mathcal{Q}$ denotes the prior input, which is $\mathcal{Z}_1$ or $\mathcal{Z}$.

adjustments without changing its resolution. Then, a $(r_1+4)\times(r_1+4)$ convolution with a stride of r_1 is used to complete the downsampling operation. To perform $\uparrow^r_{\theta^{(k)}}(\cdot)$, we first use a deconvolution with a kernel size of $(r_1+4)\times(r_1+4)$ and a stride of r_1. Then, the bilinear algorithm is used to scale the spatial resolution to $H\times W$. Finally, we use a 5×5 deconvolution with a stride of 1 for making dynamic adjustments.

Although the stratified sampling structures can better depict the nonlinear observation models and increase the depth of the network, the loss of information in multilayer transmissions may cause spatial and spectral distortions and the bilinear algorithm would create additional noise. To overcome these problems, we can treat $\downarrow^r_{\theta^{(k)}}(\cdot)$ as the feature extraction block and $\uparrow^r_{\theta^{(k)}}(\cdot)$ as the image reconstruction block. Inspired by the pyramid CNN [39] and residual CNN [54], we add the residual learning to $\uparrow^r_{\theta^{(k)}}(\cdot)$ and introduce $\mathbf{Z}$ and its downsampling version $\mathbf{Z}_1$ as the priors. Specifically, we first let $\mathbf{Z}$ to pass $\downarrow^r_{\theta^{(k)}}(\cdot)$ to obtain $\mathbf{Z}_1$. Then, we insert a three-layer convolution residual block (R-B) between two adjacent sampling layers of $\uparrow^r_{\theta^{(k)}}(\cdot)$, in which $\mathbf{Z}_1$ and $\mathbf{Z}$ are injected as priors. The residual learning can preserve the low-frequency information, better learn the high-frequency details, and dilute the effect of noise. The use of HR-MSI in different scales can further supplement the spatial information. As a result, the distortion of information can be reduced, and the quality of fusion can be improved. Figs. 3 and 4 show the detailed structures of $\downarrow^r_{\theta^{(k)}}(\cdot)$ and $\uparrow^r_{\theta^{(k)}}(\cdot)$ for r_2 as an integer and a noninteger, respectively. Equation (35) can be easily solved by a 1×1 convolution where the kernel is reshaped by $\mathbf{P}$. Equation (36) expresses a linear fusion structure where the temporary HSI $\mathcal{T}^{(k)}$ is combined with the synthetic HSI $\mathcal{E}_3{}^{(k)}$, the multiplier $\mathcal{H}^{(k)}$, and the residual information $\mathcal{E}_2{}^{(k)}$ learned from $\mathcal{Y}$ to update itself. Also, the constant parameters b_0–b_3 can be learned adaptively in the network. The detailed structure of the module $\mathcal{T}$ is shown in Fig. 5.

3) Data Module $\mathcal{A}$: Equation (18) can be divided into the following fundamental operations:

$$\mathcal{F}_1{}^{(k)} = \mathcal{V}^{(k+1)} - c_0\mathcal{G}^{(k)} \tag{39}$$
$$\mathcal{F}_2{}^{(k)} = \text{sign}(\mathcal{F}_1{}^{(k)}) \tag{40}$$
$$\mathcal{F}_3{}^{(k)} = \mathcal{F}_2{}^{(k)} \times \mathcal{F}_1{}^{(k)} - c_1 \tag{41}$$
$$\mathcal{F}_4{}^{(k)} = \max(\mathcal{F}_3{}^{(k)}, 0) \tag{42}$$
$$\mathcal{A}^{(k+1)} = \mathcal{F}_2{}^{(k)} \times \mathcal{F}_4{}^{(k)}. \tag{43}$$

Equation (39) expresses a linear fusion structure, which can be solved easily. Equations (40)–(43) represent the soft threshold algorithm, which aims to limit the learning range of the coefficients of $\mathcal{A}^{(k)}$ to ensure its sparsity. The detailed structure of module $\mathcal{A}$ is shown in Fig. 6.

4) Multiplier Module $\mathcal{G}$: Equation (10) can be divided into the following two operations:

$$\mathcal{I}_1{}^{(k)} = \mathcal{A}^{(k+1)} - \mathcal{V}^{(k+1)} \tag{44}$$
$$\mathcal{G}^{(k+1)} = \mathcal{G}^{(k)} + d_0\mathcal{I}_1{}^{(k)}. \tag{45}$$

Equation (44) can be easily solved by calculating the error between the desired coefficients $\mathcal{A}^{(k+1)}$ and the temporary coefficients $\mathcal{V}^{(k+1)}$. Equation (45) utilizes the error as the gradient to update the multiplier $\mathcal{G}^{(k)}$ with a step size of d_0. The architecture of this module is shown in Fig. 7(a).

5) Multiplier Module $\mathcal{H}$: Equation (11) can be divided into the following two operations:

$$\mathcal{I}_2{}^{(k)} = \mathcal{V}^{(k+1)} \times_3 \mathbf{P} - \mathcal{T}^{(k+1)} \tag{46}$$
$$\mathcal{H}^{(k+1)} = \mathcal{H}^{(k)} + d_1\mathcal{I}_2{}^{(k)}. \tag{47}$$

As in above, (46) can be transformed into a 1×1 convolution with the reshaped $\mathbf{P}$ as the kernel, and then, an error

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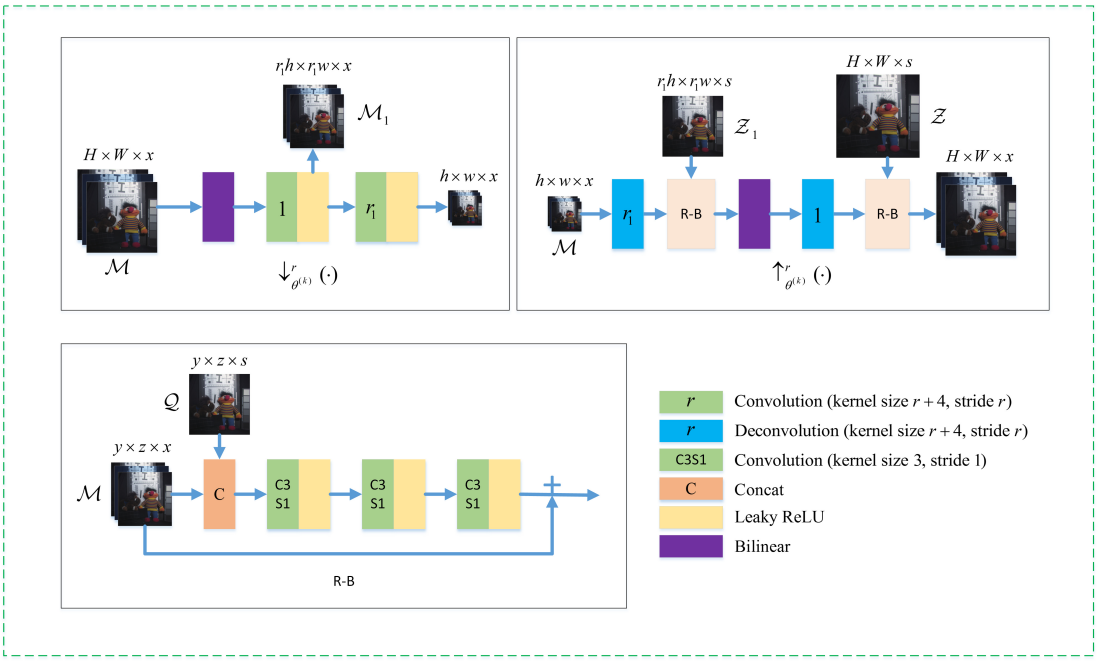

Fig. 4. Structures of the $\downarrow^r_{\theta^{(k)}}(\cdot)$ and $\uparrow^r_{\theta^{(k)}}(\cdot)$ over a decimal factor r_2. $\mathcal{M}$ denotes the current input of each block and $\mathcal{Q}$ denotes the prior input, which is $\mathcal{Z}_1$ or $\mathcal{Z}$.

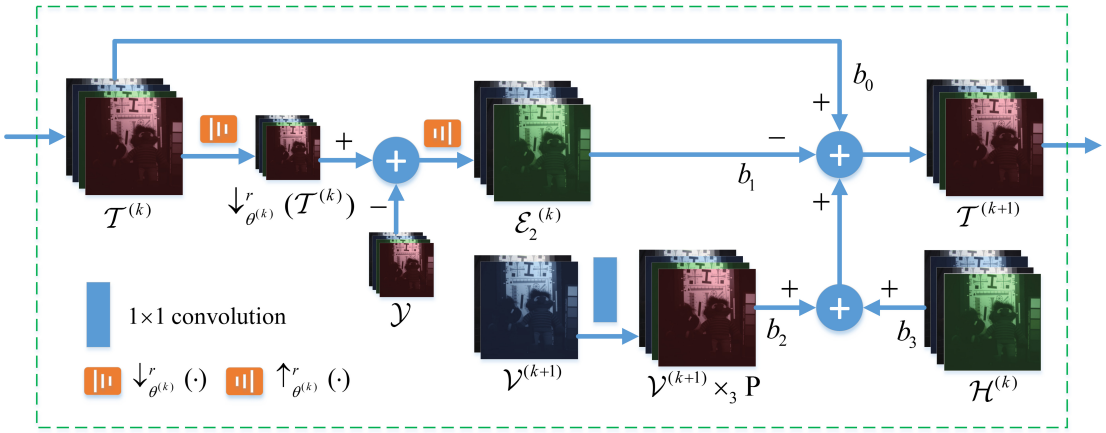

Fig. 5. Architecture of the module $\mathcal{T}$ at the kth stage.

can also be calculated. Equation (47) utilizes the error as the gradient to update the multiplier $\mathcal{H}^{(k)}$ with the step size of d_1. The architecture of this module is shown in Fig. 7(b).

6) Information Supplement Module $\mathcal{X}$: This module contributes in the compensation for approximating error in each stage, which can further improve the fusion performance. Equation (26) can be divided into the following fundamental operations:

$$\mathcal{J}_1^{(k+1)} = \mathcal{A}^{(k+1)} \times_3 \mathbf{P} \tag{48}$$

$$\mathcal{J}_2^{(k+1)} = \mathcal{V}^{(k+1)} \times_3 \mathbf{P} \tag{49}$$

$$\mathcal{J}_3^{(k+1)} = \uparrow^r_{\theta^{(k)}}(\downarrow^r_{\theta^{(k)}}(\mathcal{T}^{(k+1)}) - \mathcal{Y}) \tag{50}$$

$$\mathcal{J}_4^{(k+1)} = (\mathcal{T}^{(k+1)} \times_3 \mathbf{R} - \mathcal{Z}) \times_3 \mathbf{R}^T \tag{51}$$

$$\mathcal{X}^{(k+1)} = e_0\mathcal{T}^{(k+1)} + e_1\mathcal{J}_1^{(k+1)} + e_2\mathcal{J}^{(k+1)} - e_3\mathcal{J}_3^{(k+1)} - e_4\mathcal{J}_4^{(k+1)}. \tag{52}$$

Equations (48) and (49) can be solved easily by a 1×1 convolution with the reshaped $\mathbf{P}$ as the kernel. Equation (50) can be solved by using $\downarrow^r_{\theta^{(k)}}(\cdot)$ and $\uparrow^r_{\theta^{(k)}}(\cdot)$ as designed above. Equation (51) can be solved by the aforementioned convolutional implementation of $\mathbf{R}$ and its transpose. Equation (52) expresses a linear fusion structure, where the first three items correspond to the low-frequency information, while the last two items correspond to the high-frequency details. The architecture of the module is shown in Fig. 8. By analyzing the architecture, we can find that the desired image is fused by the main information represented by the spectral basis and the residual information generated by the simulated spectral and spatial degradation processes.

7) Dense Fusion: As we know, deep CNNs extract different features at different levels. The use of simply low- or high-level features is bound to ignore a lot of information.

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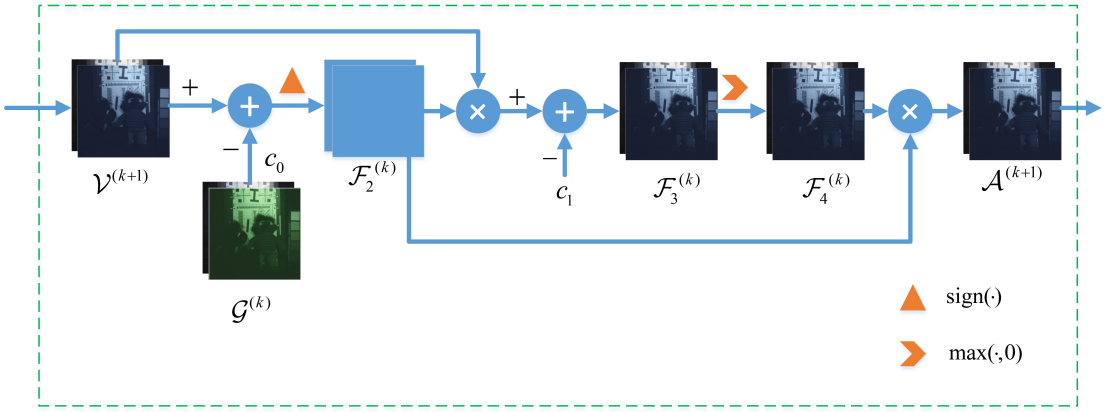

Fig. 6. Architecture of the module $\mathcal{A}$ at the kth stage.

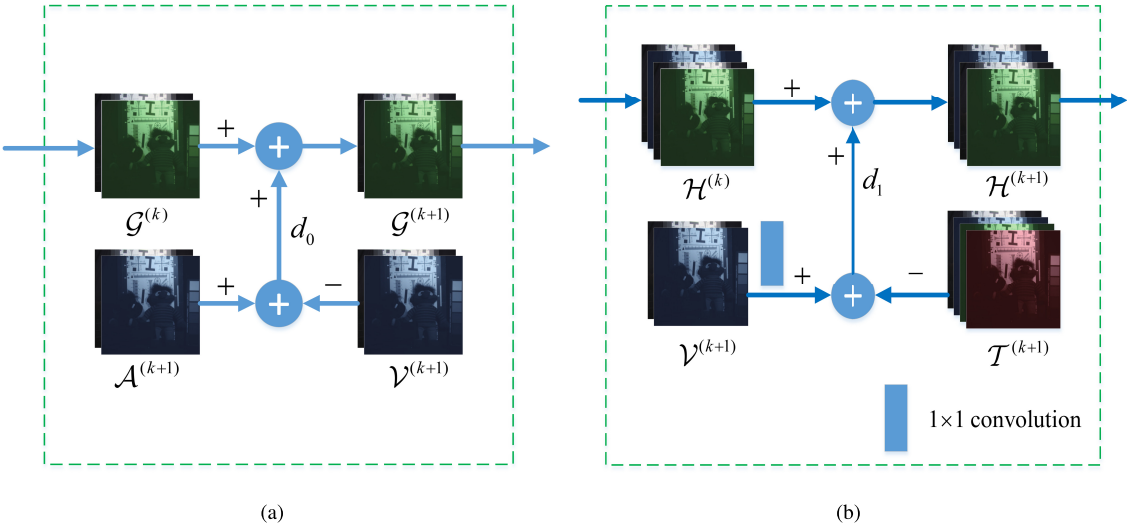

Fig. 7. Architectures of the multiplier modules at the kth stage. (a) Module $\mathcal{G}$. (b) Module $\mathcal{H}$.

A combination of different levels of features is helpful to the task of image reconstruction. In other words, the latent HSIs generated in different stages can be integrated for improving the fusion performance as they contain different pieces of spatial and spectral information. Therefore, we add the DF in our network, which concatenates the output of each stage and passes a 3×3 convolution to obtain the final HSI. The effectiveness of the DF was verified in [47].

Remark: In each stage, the network first updates the temporary spectral coefficient $\mathcal{V}^{(k)}$. Then, $\mathcal{V}^{(k)}$ is combined with the known spectral basis $\mathbf{P}$ and other constraints to form the temporary HSI $\mathcal{T}^{(k)}$. Furthermore, the desired spectral coefficient $\mathcal{A}^{(k)}$ is also generated. The multiplier $\mathcal{G}^{(k)}$ depicts the residual between $\mathcal{V}^{(k)}$ and $\mathcal{A}^{(k)}$, and it aims to keep these two coefficients infinitely close. Similarly, the multiplier $\mathcal{H}^{(k)}$ constrains $\mathcal{T}^{(k)}$ and HSI directly synthesized by $\mathbf{P}$ and $\mathcal{V}^{(k)}$ to be infinitely close. To obtain the desired HSI $\mathcal{X}^{(k)}$, the main information and the residual information are linearly fused. The main information comes from the temporary HSI $\mathcal{T}^{(k)}$ and the two HSIs represented by $\mathcal{V}^{(k)}$ and $\mathcal{A}^{(k)}$ with $\mathbf{P}$. The residual information is learned from the observation images according to the simulated degradation processes. After multiple stages, the residual information becomes more sparse and the desired HSI goes closer to the ground truth. For further improvement of the performance, the desired HSIs generated in different stages are fused to obtain the final fusion result. Each module of the network has its own physical meaning, and different modules are closely correlated, which makes the network more interpretable and facilitates the knowledge of inherent working mechanisms.

F. Network Training

1) Loss Function: Considering that L1 loss better preserves the edges and details, we adopt it to train our network. It is defined as

$$\mathcal{L} = \|\mathbf{F} - \mathbf{X}_{\text{gt}}\|_1 \tag{53}$$

where $\mathbf{F}$ and $\mathbf{X}_{\text{gt}}$ denote the fusion result and the ground truth, respectively.

2) Variable Initialization: Tensors $\mathcal{V}$, $\mathcal{T}$, $\mathcal{A}$, $\mathcal{G}$, and $\mathcal{H}$ can all be initialized to $\mathbf{0}$. To speed up the training, we intentionally initialize $\mathcal{V}$ as $\uparrow_{\theta^{(0)}}^{r}(\mathcal{Y}) \times_3 \mathbf{P}^T$, $\mathcal{T}$ as $\mathcal{Z} \times_3 \mathbf{R}^T$ and the others as $\mathbf{0}$. The constant parameters, such as a_0, are initialized to

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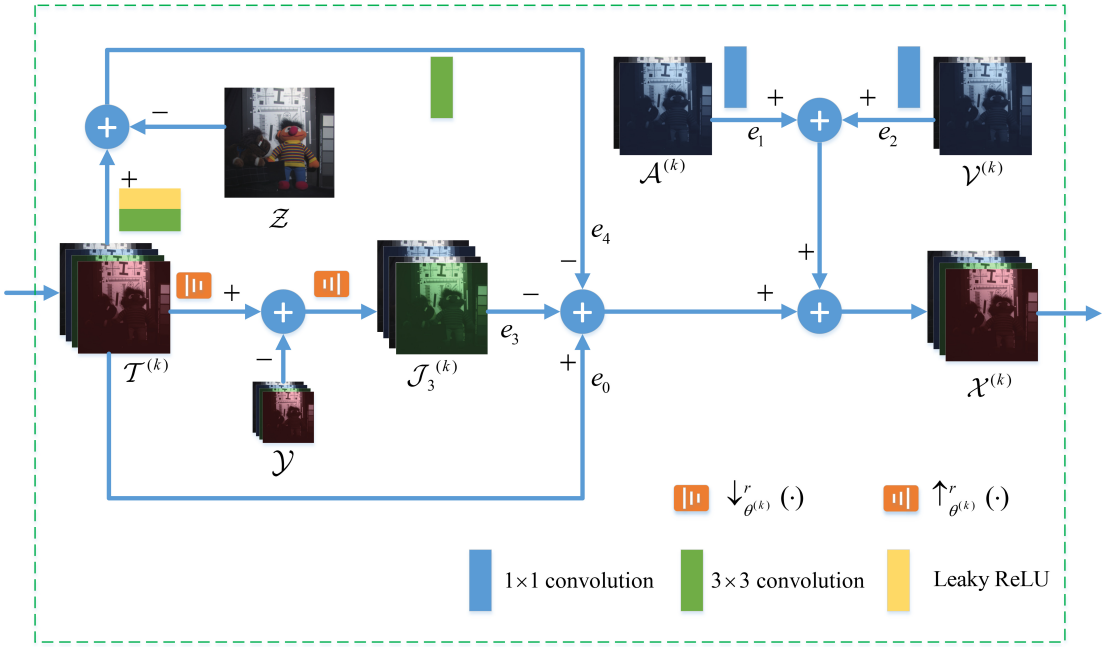

Fig. 8. Architecture of the module $\mathcal{X}$ at the kth stage.

random values by obeying a truncated normal distribution, where the mean is 0 and the standard deviation is 0.1. It should be noted that the model parameters are not shared across different stages because of the limited capacity and performance of the network.

IV. Experiments

A. Data and Experimental Setup

Two simulated datasets are used for evaluating the performance of our proposed ADMM-HFNet method, namely CAVE dataset [55] and Harvard dataset [56]. For experiments with real data, we use the University of Houston (UH) dataset.[1]

The CAVE dataset comprises 32 images, where each image has 512 × 512 pixels and 31 bands. The range of wavelength is from 400 to 700 nm. In general, the first 20 images are chosen to constitute the training set, and the remaining to constitute the test set.

The Harvard dataset consists of 50 images, where each HSI contains 31 bands (ranging from 420 to 720 nm) and has a spatial resolution of 1040 × 1392. In general, we choose the first 30 images to constitute the training set, and the remaining are for testing.

The UH dataset contains a 48-band LR-HSI of spatial size of 1202 × 4172 and an RGB image of spatial size of 24 040 × 83 440. For convenience, we crop five nonoverlapping images for experimenting. The spatial sizes of the cropped images from the HR-MSI and LR-HSI are 600 × 800 and 30 × 40, respectively. Due to the lack of reference images, the training data are constructed according to Wald's protocol. Specifically, we first downsample the RGB image 20 times with the filter estimated by HySure [26]. Then, we apply the same data argumentation techniques, including geometric flipping and rotation to both the LR-HSI and downsampled RGB image to generate 20 × 20 pairs of images. Finally, the generated LR-HSIs and RGB images are concatenated along the spatial directions to construct the training reference image and HR-MSI, respectively. The corresponding LR-HSI is generated by downsampling the reference image also with the estimated filter. During testing, the original RGB image and LR-HSI are still taken as the input for the fusion task.

For both the simulated datasets, we obtain the LR-HSIs by blurring the HR-HSIs with an 8 × 8 Gaussian filter that has a mean of 0 and a standard deviation of 2 and by downsampling every 8 pixels. The HR-MSIs are simulated via spectral degradation, where the SRF is derived from a Nikon D700 camera.[2] This SRF describes the relationship between wavelength and R-G-B bands. We select the response points having the same wavelength range as the HSI (CAVE: 400–700 nm and Harvard: 420–720 nm) and use the spline interpolation to match the number of selected points to the number of bands of the image. After the standardization by making the sum of the response values in each band to 1, the spectral downsampling matrix $\mathbf{R}$ can be formed. To verify the effectiveness under noise, a Gaussian noise is added to the LR-HSI and HR-MSI with signal-to-noise ratios (SNRs) of 30 and 40 dB, respectively.

B. Implementation Details

The Adam optimizer [57] with its default parameters is chosen to minimize the loss function for the training. The size of the HSI patch for both CAVE and Harvard datasets is set to 64 × 64, while it is 60 × 60 for the UH dataset. For constructing patches, the stride sizes in both the horizontal

[1]https://site.ieee.org/france-grss/2018/03/15/data-fusion-contest-2018-test-data-are-available/

[2]https://www.maxmax.com/spectral_response.htm

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and vertical directions are set as 16 for the two simulated datasets, while 10 for the UH dataset. The mini-batch size is 16. The total number of iterations for training is set to 160 000, and the learning rate is initialized as 4×10^{-4}, which decays by a factor of 0.5 after 20 000, 40 000, and 80 000 iterations. We train the network with Tensorflow [58] in a single GeForce GTX1660 Ti GPU. It takes about one day to finish the training.

C. Comparison of Methods

The proposed ADMM-HFNet method is compared with recent advanced fusion approaches: HySure [26], optimized twin dictionaries (OTDs) [53], image fusion by CNN denoiser (CNN-Fus) [59], deep hyperspectral image sharpening (DHSIS) [40], multispectral and hyperspectral fusion net (MHF-net) [46], dual-attention residual network (DARN) [41], and deep blind iterative fusion network (DBIN) [47]. According to the use of any prior knowledge other than LR-HSI and HR-MSI, these fusion methods can be divided into nonblind and blind methods. It is clear that HySure, MHF-net, DARN, DBIN, and our proposed ADMM-HFNet methods are blind, while OTD, CNN-Fus, and DHSIS methods are nonblind as they require additional degradation information. For a fair comparison, all the competing methods are tuned to their optimal performances. Specifically, for the HySure method, we set the number of nontruncated singular vectors as 10 for the two simulated datasets, while it is set as 4 for UH. The blurring kernel has a size of $\lfloor r \rfloor \times \lfloor r \rfloor$ in all the cases. For the OTD method, the patch size is 8×8, and both T_s and T_p (the maximum numbers of iterations) are set to 10. For the CNN-Fus method, the number of iterations is 15, the rank of the subspace is 10, and the trade-off parameter λ is 10^{-3}. For the DHSIS method, both the regularization parameters η_1 and η_2 are set as 5×10^{-4}. For the MHF-net method, the number of stages is 20, the patch size is 64×64, the batch size is 16, and the total number of training epochs is 31. For the DARN method, the number of attention blocks N is 5, the size of the training patch is 32×32, and the batch size is 16. For the DBIN method, the number of stages is 5, the patch size is 64×64, the batch size is 16, and the total number of training iterations is 250 000. The other parameters take the default values as reported in the original articles.

In the quantitative comparison, the peak SNR (PSNR) [60], spectral angle mapper (SAM) index [61], root-mean-square error (RMSE) [62], structural similarity (SSIM) index [63], erreur relative globale adimensionnelle de synthèse (ERGAS) [64], and universal image quality index (UIQI) [65] are used for evaluation. PSNR, RMSE, and SSIM measure the fusion quality in the spatial domain. SAM measures the spectral quality, which calculates the average spectral angular between the fused and reference images. ERGAS and UIQI measure the overall quality of the fusion result, including space and spectrum. Except for SAM and RMSE, all other metrics are extended from single-band images to HSIs by averaging all bands.

D. Parameter Selection

In our proposed method, two key parameters influence the fusion quality: the number of stages K and the rank c of the spectral basis. Here, analyzing the effects of these two parameters on fusion, we select more appropriate values for better performance. Since the CAVE and Harvard datasets have the same number of bands and similar wavelength ranges, we consider that they can share the same K and c also. Therefore, for convenience, experiments for selecting parameters can be performed only on the CAVE dataset.

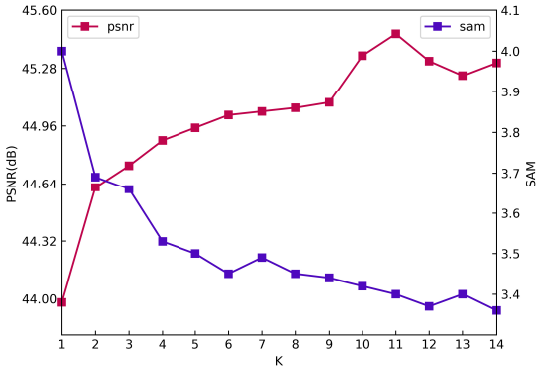

Fig. 9. Effect of parameter K on the proposed method for the CAVE dataset.

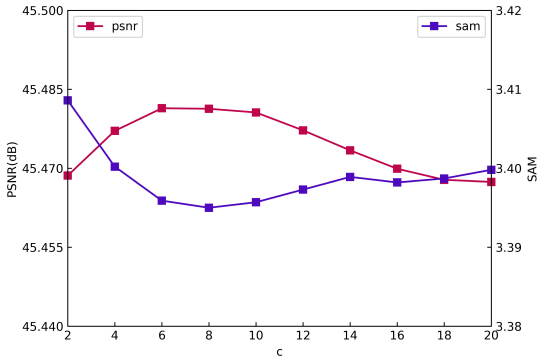

Fig. 10. Effect of parameter c on the proposed method for the CAVE dataset.

The parameter K controls the depth of the deep network, which should be set properly to ensure good performance and kept as minimal as possible. Fig. 9 shows the variation of PSNR and SAM of reconstructed images with K. We can see that PSNR keeps rising when K varies from 1 to 11, and then, it falls slightly. Similarly, SAM maintains an overall downward trend with increasing K. Thus, we set K as 11 for obtaining the optimal performance.

The parameter c controls the amount of reduction in dimensionality, the proper value of which can denoise and improve the fusion performance. To analyze the effect of this parameter, the PSNR and SAM as functions of c are presented in Fig. 10. We can see that PSNR first rises with c increasing from 2 to 8, and then, it falls. Similarly, SAM decreases when c ranges from 2 to 8, and then, it keeps an overall increasing trend, which indicates that the proper value of c is 8 and the spectral signatures actually appear in a low-dimensional subspace. Therefore, we set the rank of the spectral basis as 8 for obtaining the best performance.

E. Performance Comparison

1) Performance Comparison With Simulated Datasets: The quantitative results, in terms of PSNR, SAM, SSIM, RMSE,

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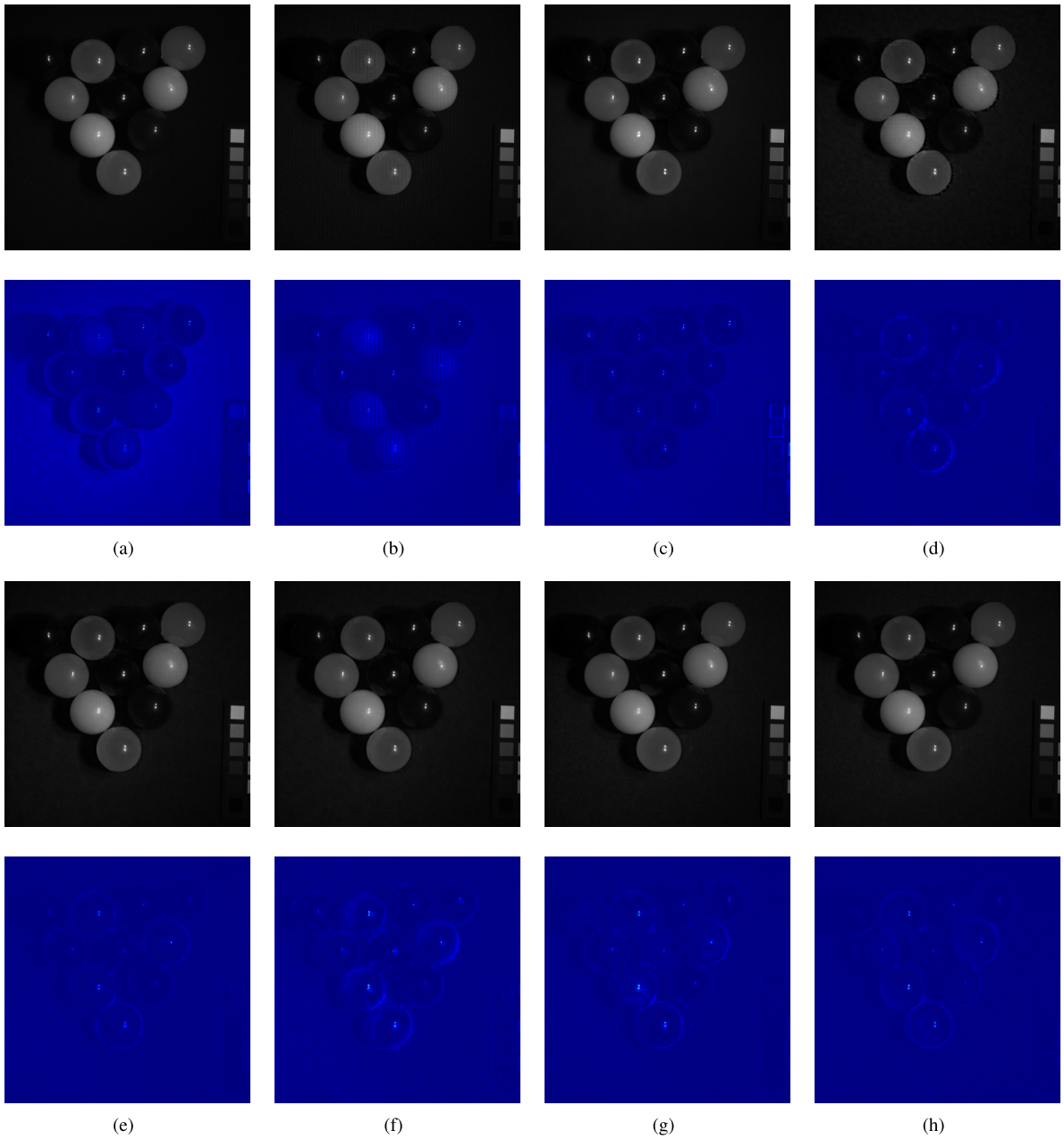

Fig. 11. Visual results of the CAVE dataset at band 31. (a) HySure. (b) OTD. (c) CNN-Fus. (d) DHSIS. (e) MHF-net. (f) DARN. (g) DBIN. (h) ADMM-HFNet. First row: reconstructed images. Second row: reconstructed errors.

ERGAS, and UIQI, of the competing methods on the CAVE dataset are shown in Table II. For clarity, the optimal results are marked in red color, and the second best ones are in blue color. We can see in Table II that our proposed method and the DBIN method perform better than other competing methods. It means that our method can better improve the spatial resolution while preserving the spectral information also. The reconstructed images at the 31st band and the corresponding reconstructed errors are shown in Fig. 11. The reconstructed errors measure the difference between the reconstructed and reference images. It can be observed that our method and the MHF-net method reconstruct the details better than other competing methods. Fig. 12(a) compares the PSNR performance of the competing methods at different spectral bands. It can be observed that, among the competing methods, the performance of our ADMM-HFNet is the best at most of the spectral bands.

The averaged quantitative results for the Harvard dataset are shown in Table III. The experimental results demonstrate that our method achieves the most outstanding performance in all aspects. As we can see in Fig. 13, our method and DBIN contain lesser reconstruction error and have clearer details. Fig. 12(b) shows the average PSNR curves for the Harvard dataset as functions of the spectral bands, where we can see that DBIN and our method lead in most of the spectral bands.

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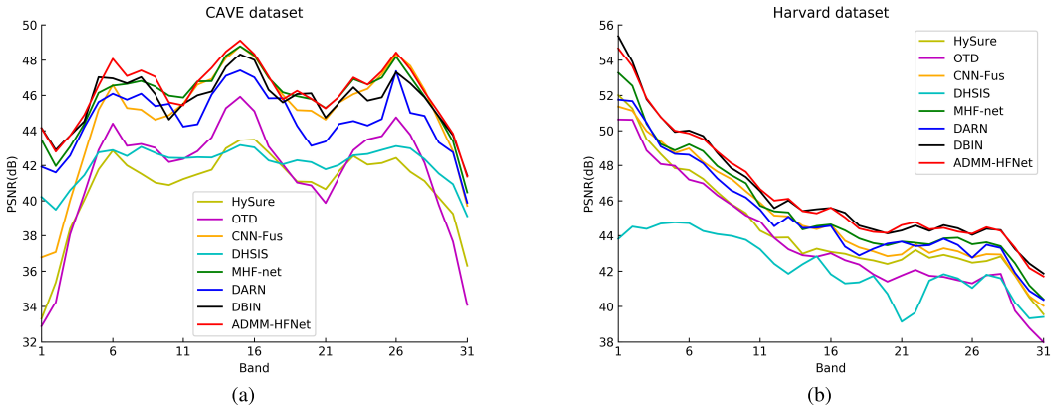

Fig. 12. Average PSNR curves as functions of the spectral bands. (a) CAVE dataset. (b) Harvard dataset.

TABLE II
AVERAGED QUANTITATIVE RESULTS FOR CAVE DATASET

Methods	PSNR	SAM	SSIM	RMSE	ERGAS	UIQI
Best Values	$+\infty$	0	1	0	0	1
HySure	39.69	10.04	0.9711	0.0109	1.3862	0.9518
OTD	39.52	8.50	0.9758	0.0113	1.3112	0.9646
CNN-Fus	43.40	5.34	0.9850	0.0071	0.8771	0.9789
DHSIS	41.85	9.80	0.9568	0.0086	1.0270	0.9221
MHF-net	45.23	4.88	0.9878	0.0059	0.7064	0.9810
DARN	44.02	4.47	0.9889	0.0067	0.7954	0.9861
DBIN	45.02	3.38	0.9922	0.0062	0.7400	0.9930
ADMM-HFNet	45.48	3.39	0.9925	0.0058	0.7068	0.9925

TABLE III
AVERAGED QUANTITATIVE RESULTS FOR HARVARD DATASET

Methods	PSNR	SAM	SSIM	RMSE	ERGAS	UIQI
Best Values	$+\infty$	0	1	0	0	1
HySure	43.41	3.77	0.9790	0.0082	1.5523	0.9856
OTD	42.21	4.81	0.9658	0.0151	2.1420	0.9802
CNN-Fus	43.89	4.39	0.9779	0.0078	1.4576	0.9853
DHSIS	41.71	6.53	0.9586	0.0097	2.3565	0.9546
MHF-net	44.50	3.68	0.9812	0.0074	1.2112	0.9914
DARN	44.10	3.75	0.9806	0.0078	1.4164	0.9892
DBIN	45.33	3.04	0.9834	0.0070	1.0851	0.9950
ADMM-HFNet	45.53	3.04	0.9834	0.0069	1.0799	0.9952

2) Performance Comparison With Real Datasets: To further illustrate the effectiveness of our proposed method in real fusion scenarios, we present the fusion results for the UH dataset. Here, also, the number of stages K is set as 11. In this experiment, we empirically set the rank of the spectral basis c as 4. Since OTD, CNN-Fus, and DHSIS are nonblind methods that require real SRF and PSF, in this case, we compare only the blind methods. Fig. 14 shows the false-color images of a reconstructed area, where R-G-B corresponds to the 46-21-6 bands of reconstructed images. We can observe that DBIN and our method give better visual effects with clearer details and they are much closer to the real LR-HSI in terms of color and brightness.

TABLE IV
EFFECTIVENESS OF THE INFORMATION MODEL ON THE CAVE DATASET

Methods	PSNR	SAM	SSIM	RMSE	ERGAS	UIQI
Best Values	$+\infty$	0	1	0	0	1
NOT	44.11	3.92	0.9910	0.0068	0.8082	0.9898
ADD	45.48	3.39	0.9925	0.0058	0.7068	0.9925

TABLE V
FUSION UNDER THE NONINTEGER RESOLUTION RATIO ON THE CAVE DATASET

Methods	PSNR	SAM	SSIM	RMSE	ERGAS	UIQI
Best Values	$+\infty$	0	1	0	0	1
DARN	45.22	4.15	0.9899	0.0058	0.7084	0.9866
ADMM-HFNet	46.01	3.27	0.9926	0.0055	0.6607	0.9926

F. Effectiveness of the Information Supplement Model

In this article, we have proposed an information supplement model and embedded its solver in the deep network to supplement the information that cannot be represented by the spectral basis. To illustrate its effectiveness, we compare the performances with and without this model, which are denoted as "ADD" and "NOT," respectively. Experimental results for the CAVE dataset are shown in Table IV. It can be observed that the addition of this model could significantly improve the fusion performance in all the quantitative terms. Therefore, it can be concluded that the proposed information supplement model is effective.

G. Fusion Under the Noninteger Resolution Ratio

The proposed hierarchical sampling structure can handle fusion with arbitrary resolution ratios. We verify its effectiveness and performance by carrying out a fusion experiment under a noninteger resolution ratio. The experiment is conducted on the CAVE dataset. We simulate the HR-MSI as above, while the LR-HSI is simulated by using a 5×5

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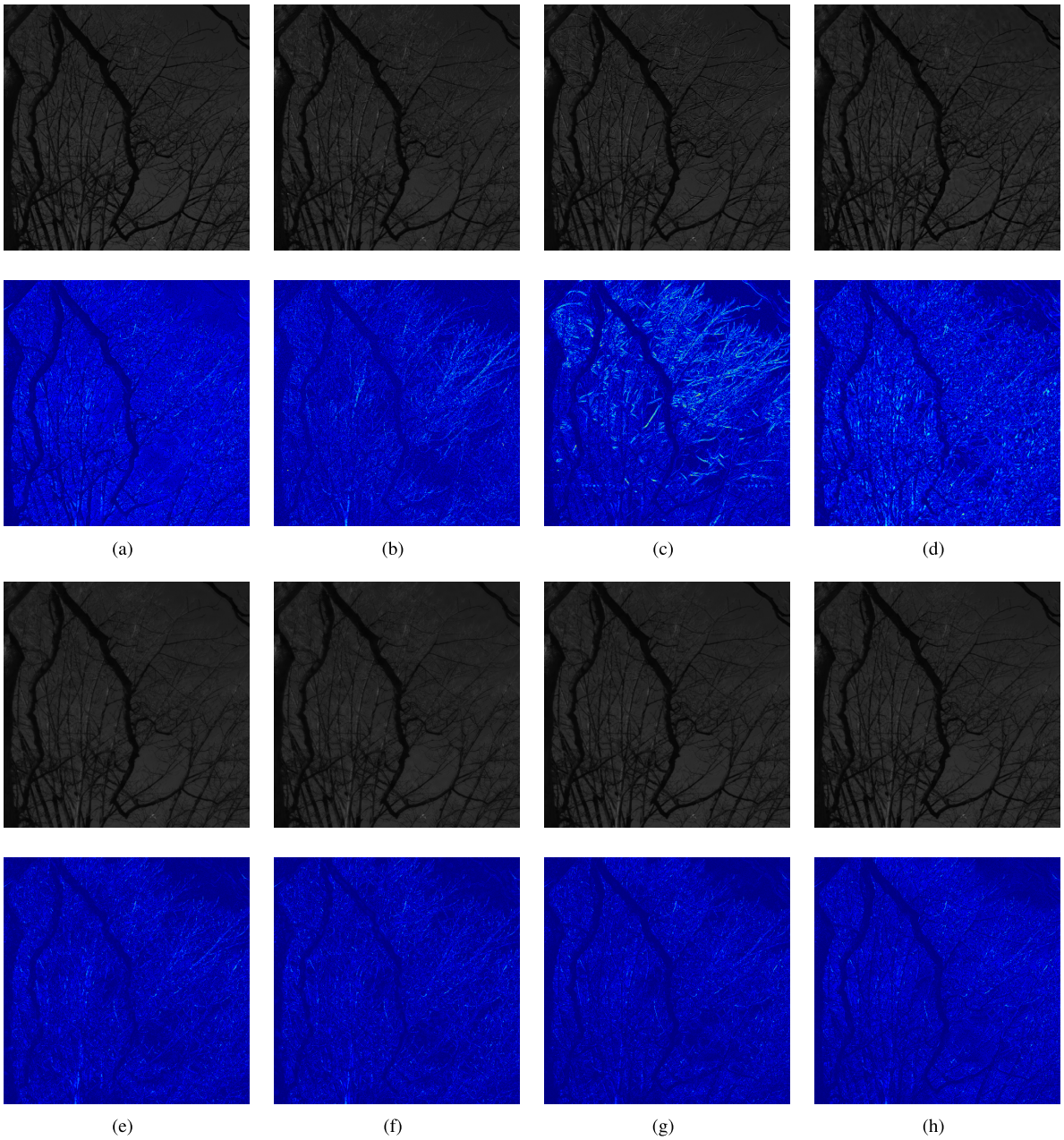

Fig. 13. Visual results of Harvard dataset at band 31. (a) HySure. (b) OTD. (c) CNN-Fus. (d) DHSIS. (e) MHF-net. (f) DARN. (g) DBIN. (h) ADMM-HFNet. First row: reconstructed images. Second row: reconstructed errors.

Gaussian filter with a standard deviation of 2 and then by downsampling the spatial resolution to 93 $\times$ 93 with the nearest algorithm. Hence, the resolution ratio is 5.5. Since none of the compared methods except DARN can directly deal with the fusion of noninteger resolution ratios, we compare our method with DARN only.

The averaged quantitative results are shown in Table V. We can find that our ADMM-HFNet still achieves an outstanding performance under the decimal resolution ratio, which demonstrates its robustness to a certain extent.

H. Computational Efficiency

The comparison of running time of our proposed ADMM-HFNet with those of other competing methods on the three datasets is shown in Table VI. All the experiments are performed through Python 3.6 or MATLAB R2016a in an Intel Core i5-9300H CPU with 2.40 GHz and 8-GB RAM. As can be seen from the average time, DBIN and our method are faster than other competing methods. This advantage has been achieved as our method comes mainly from a model-driven network and matrix decomposition. The model-driven network can greatly simplify a network and effectively improve its performance. The matrix decomposition can largely reduce the dimensionality of HSI data and thus save a lot of calculations. The DARN method suffers from a lot of computational time as it needs first to learn an initial upsampled image from the noise via a deep hyperspectral prior approach in an unsupervised way.

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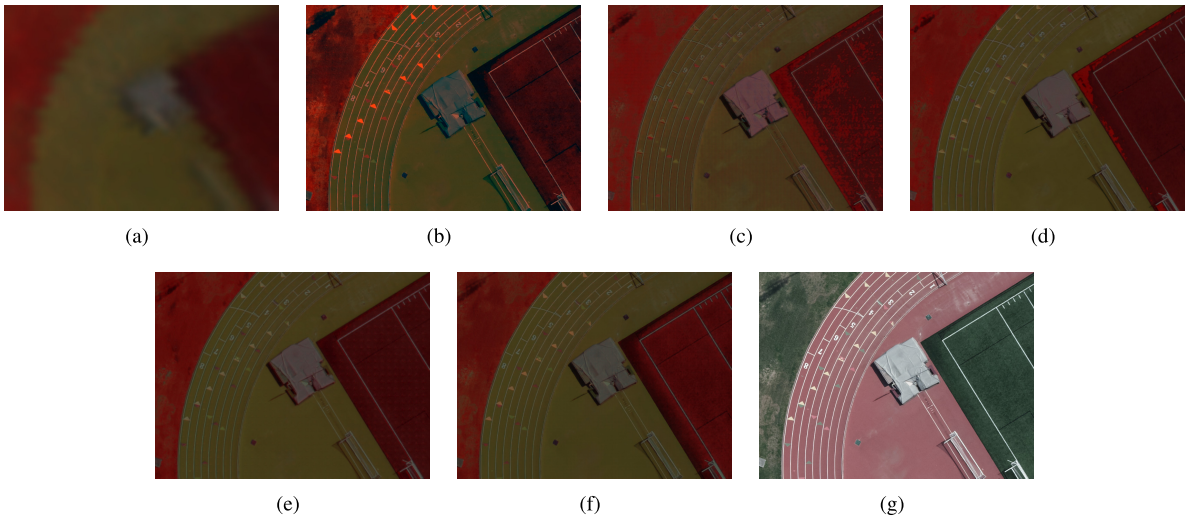

Fig. 14. Visual results of UH dataset. We show the reproduced false-color images obtained from the reconstructed HSIs with bands 46-21-6 as R-G-B. (a) LR-HSI. (b) HySure. (c) MHF-net. (d) DARN. (e) DBIN. (f) ADMM-HFNet. (g) HR-MSI.

TABLE VI
COMPARISON OF RUNNING TIME (SECOND)

Methods	CAVE	Harvard	UH	Average
HySure	186.01	1136.1	180.76	500.96
OTD	122.98	606.02	-	364.50
CNN-Fus	74.755	90.675	-	82.715
DHSIS	6.2370	47.206	-	26.722
MHF-net	2.9130	6.1930	16.968	8.6913
DARN	3597.6	18961	7352.1	9970.2
DBIN	2.0510	8.5860	10.927	7.1880
ADMM-HFNet	2.0440	5.7230	16.048	7.9383

V. CONCLUSION

In this article, a model-driven deep approach based on matrix decomposition is proposed to address the LR-HSI and HR-MSI fusion problems. The desired HSI is decomposed into spectral basis and coefficients. The spectral basis is estimated from LR-HSI via SVD as it shares almost the same spectral information as the desired HSI. To learn the coefficients, we construct a model that considers observation models and sparsity of the coefficients. Instead of directly solving the model via the ADMM technique, we unfold its iterations into an optimization-based deep network. The regularization parameters as well as the SRF and PSF parameters can be learned adaptively. Considering that some information cannot be learned from a linear unmixing model, we construct another model to supplement it. The information supplement model can also be embedded in the deep network. Experiments with three datasets have proved the superiority of the proposed ADMM-HFNet in both qualitative and quantitative aspects. Besides, it can handle the fusion task of arbitrary resolution ratios and shows its robustness to a certain degree.

The proposed model-driven deep learning approach is supposed to achieve promising performance. It can be migrated to other image processing tasks, such as classification and detection. In future work, we will attempt to learn both the spectral basis and the coefficients in the deep network so that the entire network can be made to be a purely end-to-end system.

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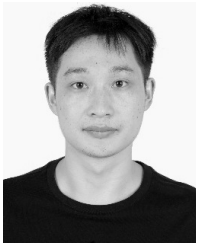

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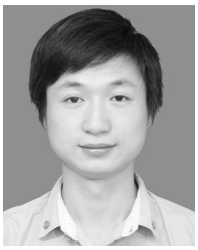

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